

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Rising Waters– AI-Powered Flood Prediction & Early Warning System
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses historical datasets only; no real-time weather or river-level data.	Prediction accuracy depends on static data.	High
2	Supports a single Flask server with limited concurrent users.	Performance may decrease with increased user traffic.	Medium
3	Basic authentication and local storage without cloud deployment.	Limited security, scalability, and accessibility.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user Flask application.	Deploy on cloud platforms (AWS, Azure, Render) with load balancing to support multiple users.
2	Data Storage	Local CSV dataset and SQLite storage.	Migrate to PostgreSQL or MongoDB with cloud storage for large-scale data management.
3	Performance	Offline ML prediction using a pre-trained model.	Integrate real-time weather APIs, optimize ML models, and enable faster inference.
4	Security	Basic user authentication.	Implement JWT authentication, HTTPS, role-based access control, and encrypted data storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time weather and river-level APIs for live flood prediction.	August 2026	Improves prediction accuracy with real-time environmental data.

Phase 3	Develop a mobile application with SMS and push notification alerts.	September 2026	Enables instant flood warnings and improves accessibility.
Phase 4	Integrate IoT sensors, GIS mapping, and cloud deployment for large-scale monitoring.	October 2026	Provides a scalable, intelligent disaster management platform with real-time monitoring.

Project Vision

The current implementation demonstrates the core functionality of an AI-based flood prediction system using Machine Learning and a Flask web application. Future enhancements will transform **Rising Waters** into a comprehensive disaster management platform by integrating real-time data sources, cloud infrastructure, IoT devices, GIS visualization, and automated notification services. These improvements will enable the system to support larger user bases, deliver more accurate predictions, and provide timely alerts to communities and disaster management authorities, ultimately helping reduce the impact of floods and improve public safety.